

Q1. JEE Main 2026 (23 January Shift 1)

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Consider a ferromagnetic material :

Assertion (A) : The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains.

Reason (R): At high enough temperature, the domain structure of ferromagnetic material disintegrates. Thus, magnetization will disappear at high enough temperature known as Curie temperature.

In the light of the above statements, choose the correct answer from the options given below :

- (1) (A) is true but (R) is false
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

ANSWERS AND SOLUTIONS

1. (2)

1. (2) Assertion (A) states that individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact to form domains. This is a correct description of the domain theory of ferromagnetism, where exchange interactions lead to spontaneous alignment of moments within small regions called domains.

Reason (R) states that at high temperatures (above the Curie temperature T_C), the domain structure disintegrates due to thermal agitation, and the material becomes paramagnetic. This is also a correct statement regarding the temperature dependence of ferromagnetic materials.

However, Reason (R) describes the effect of temperature on the domain structure, whereas Assertion (A) describes the fundamental nature and formation of domains in the ferromagnetic state. The disintegration of domains at high temperature is not the reason why they form spontaneously at lower temperatures. Therefore, both statements are true, but (R) is not the correct explanation of (A).